# STAR-RIS Enhanced UAV-MEC Networks: Bi-Directional Task Offloading

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***Abstract*—This paper introduces a novel multi-user mobile edge computing (MEC) scheme facilitated by simultaneously transmitting and reflecting reconfigurable intelligent surface (STAR-RIS) and unmanned aerial vehicle (UAV). Unlike existing MEC approaches, the proposed scheme enables bi-directional offloading, allowing users to concurrently offload tasks to the MEC servers located at the ground base station (BS) and UAV with STAR-RIS support. Specifically, we formulate an optimization problem aiming at maximizing the amount of the offloaded tasks of users while ensuring the quality of service (QoS) constraints by jointly optimizing the resource allocation, user scheduling, passive beamforming of the STAR-RIS, and the UAV trajectory. A block coordinate descent (BCD) iterative algorithm designed with the successive convex approximation (SCA) technique is proposed to effectively handle the formulated non-convex optimization problem with strongly coupled variables. Simulation results indicate that the proposed STAR-RIS enhanced UAV-enabled MEC scheme possesses significant advantages in enhancing the system's computational capability over other baseline schemes including the conventional RIS-aided scheme.**

***Index Terms*—STAR-RIS, unmanned aerial vehicle (UAV), mobile edge computing (MEC).**

## I. Introduction

As for the implementation of those computation-intensive and latency-critical applications, e.g., autonomous driving, augmented and virtual reality, etc., it presents significant challenges for the widely used center-cloud computing framework to effectively handle large amounts of data in swift actions [1]. To address this challenge, the technology of mobile edge computing (MEC) has emerged as a promising solution for effectively tackling the significantly augmented computational necessity, since it is able to bring the capabilities of cloud computing to the network edge and enables data processing in close proximity. Consequently the MEC technique can perform well in reducing network congestion, and improving service quality and user experience. The benefits introduced by MEC have garnered significant attention from researchers [2]–[4].

While the MEC technology provides an effective means to enhance the network computing capabilities, the traditional placement strategy for MEC servers near the ground base stations (BSs) or access points (APs) may result in a limited service coverage. To overcome this limitation, the unmanned aerial vehicles (UAVs) are leveraged to assist the task completion of the MEC networks due to the inherent advantages of UAVs, such as exceptional mobility and flexibility [5]–[8]. Although UAV is able to effectively improve the computing capacity of MEC networks, the existing UAV-aided MEC schemes are designed by adapting to uncontrollable random wireless channels, which seriously limits the task offloading efficiency. The reconfigurable intelligent surface (RIS) technique emerges as a promising solution to address this challenge [9] and has been incorporated into various kinds of communication systems including MEC networks [10]–[13]. In particular, a RIS-assisted UAV-enabled MEC scheme considering the aim of maximizing energy efficiency is proposed in [10]. This MEC scheme utilizes the RIS positioned on the building to enhance the signals of users' task offloading and direct them towards the MEC server located on the UAV. To improve the flexibility of the RIS and the computational capacity of the MEC systems, a MEC scheme utilizing two UAVs has been introduced in [13], where one UAV is equipped with a RIS to support the transmission of task offloading signals from users to the MEC server positioned on the second UAV.

Actually, the existing RIS-assisted UAV-enabled MEC schemes has certain limitations: (i) The UAV may serve as multiple roles, which imposes a challenge on UAV hardware design. (ii) The majority of MEC schemes are not energy and time efficient (iii) It is challenging for the traditional RIS-assisted UAV-enabled MEC schemes to offload users' tasks to both the BS and the UAV simultaneously in a more efficient manner. Recently, an advanced RIS technology, named as simultaneously transmitting and reflecting reconfigurable intelligent surface (STAR-RIS), has drawn great attention from both academia and industry [14]. STAR-RIS can split the incident signal into two parts, where one part is reflected to the same side of the incident signal and the other part is transited to the opposite side, allowing a 360° coverage, compared with the conventional RIS. Due to the inherent advantages, STAR-RIS possesses an enormous application potentials in various wireless communication systems, e.g., [15]–[17]. Hence, to enhance the current MEC schemes, we try to incorporate the STAR-RIS into UAV-assisted MEC networks, allowing the simultaneous bi-directional offloading strategy for the cooperation between the UAV and the BS. In this paper, a novel MEC scheme leveraging STAR-RIS and UAV enables

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users to offload computing tasks to MEC servers at the BS and UAV bidirectionally. Then, we develop an optimization problem to maximize the system's offloaded ability while meeting users' quality of service (QoS) requirements through integrated resource allocation, user scheduling, passive beamforming, and UAV trajectory design. To address this non-convex optimization problem, a iterative algorithm designed with the successive convex approximation (SCA) technique is proposed.

## II. System Model

Fig. 1 illustrates the architecture of the UAV-enabled MEC network with the assistance of the STAR-RIS. The network comprises a ground base station (BS), $K$ users each equipped with a single antenna, a UAV equipped with a signal antenna and installed with a STAR-RIS featuring $M$ elements, along with two MEC servers situated respectively at the BS and the UAV. This study adopts the energy splitting protocol for the STAR-RIS, wherein all elements incorporated within the STAR-RIS possess the capability to simultaneously reflect (R) and transmit (T) incident signals [14]. This feature enables users to offload their computing tasks in a bidirectional manner concurrently to the MEC server located at the BS using the R mode of the STAR-RIS, and to the MEC server mounted on the UAV using the T mode.

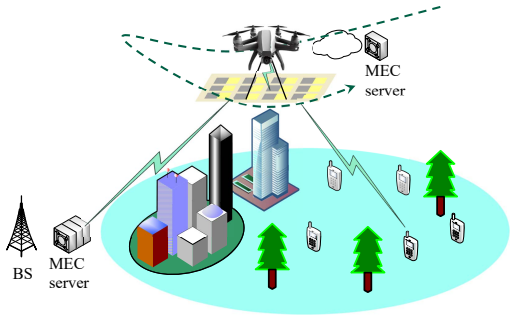



Fig. 1. The UAV-enabled MEC network with bi-directional offloading strategy supported by the STAR-RIS.

In this paper, we divide the mission period $T$ into $N$ equal time slots, i.e., $\delta_{\rm t} = T/N$, which is sufficiently small. Considering that the UAV is equipped with a transceiver that has a single antenna, the time division multiple access (TDMA) protocol is implemented to handle users' offloaded tasks. Hence, we use the variable $\zeta_k[n] \in \{0,1\}$ for $k \in \mathcal{K} \triangleq \{1,\cdots,K\}, n \in \mathcal{N} \triangleq \{1,\cdots,N\}$ to represent the user association decision for task offloading in a time slot, which needs to satisfy the following constraint:

$$\sum_{k=1}^{K} \zeta_k[n] = 1,\ \zeta_k[n] \in \{0,1\},\ \forall k \in \mathcal{K}, n \in \mathcal{N}. \tag{1}$$

In order to clearly describe the considered scenario, we assume that all nodes are situated in a 3D Cartesian coordinate system. The positions of the BS and the $k$-th user are respectively denoted as $\mathbf{q}_{\rm BS} = [x_{\rm BS}, y_{\rm BS}, z_{\rm BS}]^T$ and $\mathbf{q}_k = [x_k, y_k, 0]^T$. It is assumed that the UAV flies at a fixed altitude $H$ and its position remains constant within a given time slot considering the small value of $\delta_{\rm t}$. Consequently, the location of the UAV in the $n$-th time slot can be represented as $\mathbf{q}_{\rm ua}[n] = [x_{\rm ua}[n], y_{\rm ua}[n], H]^T,\ n \in \mathcal{N}$, which should adhere to the subsequent flight constraints:

- $\mathbf{v}_{\rm ua}[n] = \frac{\mathbf{q}_{\rm ua}[n+1]-\mathbf{q}_{\rm ua}[n]}{\delta_{\rm t}}, \|\mathbf{v}_{\rm ua}[n]\| \le v_{\max}$,
- $\mathbf{a}_{\rm ua}[n] = \frac{\mathbf{v}_{\rm ua}[n+1]-\mathbf{v}_{\rm ua}[n]}{\delta_{\rm t}}, \|\mathbf{a}_{\rm ua}[n]\| \le a_{\max}$,
- $\mathbf{q}_{\rm ua}[0] = \mathbf{q}_{\rm I}, \mathbf{q}_{\rm ua}[N+1] = \mathbf{q}_{\rm F}$,

where $\|\mathbf{v}_{\rm ua}[n]\|$ and $\|\mathbf{a}_{\rm ua}[n]\|$ respectively represent the flight speed and the acceleration of the UAV in the $n$-th time slot, with the maximum values of $v_{\max}$ and $a_{\max}$. In terms of $\mathbf{q}_{\rm I}$ and $\mathbf{q}_{\rm F}$, they can serve as the ground station where the UAV can access a reliable power supply and receive necessary maintenance.

### A. Channel Model

Due to the fact that the UAV flights at a high altitude, we assume that the line-of-sight (LoS) channels between the ground users/BS and aerial UAV/STAR-RIS can always be guaranteed. It is assumed that the STAR-RIS adopts the uniform planar array (UPA) with $M_x$ elements along $x$-axis direction and $M_y$ elements along $y$-axis direction, i.e., $M = M_x M_y$. Hence, the channel between the $k$-th user ($\varsigma = {\rm r}k$)/BS ($\varsigma = {\rm rb}$) and the STAR-RIS in the $n$-th time slot for $k \in \mathcal{K},\ n \in \mathcal{N}$ can be expressed as

$$\mathbf{h}_\varsigma[n] = \sqrt{\frac{\rho}{d_\varsigma^{\alpha_\varsigma}[n]}}\widehat{\mathbf{h}}_\varsigma[n] \in \mathbb{C}^{M\times 1},\ \varsigma \in \{{\rm r}k, {\rm rb}\}, \tag{2}$$

where

$$\begin{aligned}&\widehat{\mathbf{h}}_\varsigma[n] = \\ &\left[1,\cdots,e^{-j\frac{2\pi d}{\lambda}(m_{\rm x}-1)\xi_\varsigma[n]},\cdots,e^{-j\frac{2\pi d}{\lambda}(M_{\rm x}-1)\xi_\varsigma[n]}\right]^T \otimes \\ &\left[1,\cdots,e^{-j\frac{2\pi d}{\lambda}(m_{\rm y}-1)\chi_\varsigma[n]},\cdots,e^{-j\frac{2\pi d}{\lambda}(M_{\rm y}-1)\chi_\varsigma[n]}\right]^T,\end{aligned} \tag{3}$$

with $\rho = (\frac{\lambda}{4\pi})^2$, $\lambda$ is the wavelength of the carrier frequency, $\alpha_\varsigma$ indicating the pass loss exponent, $d_\varsigma[n]$ denoting the distance between $k$-th user/BS and the STAR-RIS. $d$ denotes the adjacent element separation of the STAR-RIS. In addition, the $\xi_\varsigma[n]$ and $\chi_\varsigma[n]$ corresponding to the $k$-th user and the BS are respectively calculated as

- $\xi_{{\rm r}k}[n] = \cos(\phi_{{\rm r}k}[n])\sin(\theta_{{\rm r}k}[n]) = \frac{x_{\rm ua}[n]-x_k}{\|\mathbf{q}_{\rm ua}[n]-\mathbf{q}_k\|}$,
- $\xi_{\rm rb}[n] = \cos(\phi_{\rm rb}[n])\sin(\theta_{\rm rb}[n]) = \frac{x_{\rm BS}-x_{\rm ua}[n]}{\|\mathbf{q}_{\rm BS}-\mathbf{q}_{\rm ua}[n]\|}$,
- $\chi_{{\rm r}k}[n] = \sin(\phi_{{\rm r}k}[n])\sin(\theta_{{\rm r}k}[n]) = \frac{y_{\rm ua}[n]-y_k}{\|\mathbf{q}_{\rm ua}[n]-\mathbf{q}_k\|}$,
- $\chi_{\rm rb}[n] = \sin(\phi_{\rm rb}[n])\sin(\theta_{\rm rb}[n]) = \frac{y_{\rm BS}-y_{\rm ua}[n]}{\|\mathbf{q}_{\rm BS}-\mathbf{q}_{\rm ua}[n]\|}$.

It is important to note that the connection between the UAV and STAR-RIS should be described as the near-field channel, denoted as $\mathbf{h}_{\rm ru}$, considering the fact that the distance between the UAV and STAR-RIS is extremely small. Thus, $\mathbf{h}_{\rm ru}$ can be expressed as [18]

$$\mathbf{h}_{\rm ru} = \boldsymbol{\alpha} \circ \mathbf{a}_{\rm ru} \in \mathbb{C}^{M\times 1}, \tag{4}$$

where $\boldsymbol{\alpha} = \left[\frac{\lambda}{4\pi r_1},\cdots,\frac{\lambda}{4\pi r_m},\cdots,\frac{\lambda}{4\pi r_M}\right]^T$, $\mathbf{a}_{\rm ru} = \left[e^{-j\frac{2\pi r_1}{\lambda}},\cdots,e^{-j\frac{2\pi r_m}{\lambda}},\cdots,e^{-j\frac{2\pi r_M}{\lambda}}\right]^T$, with $r_m$ denotes the

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distance between the antenna located on the UAV and the $m$-th element of the STAR-RIS, where $m \in \mathcal{M} \triangleq \{1, \cdots, M\}$. Notably, the channel $\mathbf{h}_{\rm ru}$ remains invariant throughout, attributed to the fixed spatial relationship maintained between the UAV and the STAR-RIS.

### B. Task Offloading and Computation Model

The offloading rates achieved by the $k$-th user in the $n$-th time slot to the BS and UAV are respectively given by

- $R_k^{\rm ua}[n] = \zeta_k[n] B \log_2\left(1 + \frac{p\left|\mathbf{h}_{\rm ru}^H \mathbf{\Theta}_{\rm t}^H[n] \mathbf{h}_{{\rm r}k}[n]\right|^2}{\sigma_{\rm ua}^2}\right)$,
- $R_k^{\rm BS}[n] = \zeta_k[n] B \log_2\left(1 + \frac{p\left|\mathbf{h}_{\rm rb}^H[n] \mathbf{\Theta}_{\rm r}^H[n] \mathbf{h}_{{\rm r}k}[n]\right|^2}{\sigma_{\rm BS}^2}\right)$,

where $p$ denotes the unified transmitted power of users, $B$ is the bandwidth of the system, $\sigma_{\rm ua}^2$ and $\sigma_{\rm BS}^2$ respectively denote the noise power at the BS and the UAV. In addition, $\mathbf{\Theta}_\kappa[n] = {\rm Diag}\left\{\beta_\kappa^1[n] e^{j\phi_\kappa^1[n]}, \cdots, \beta_\kappa^m[n] e^{j\phi_\kappa^m[n]}, \cdots, \beta_\kappa^M[n] e^{j\phi_\kappa^M[n]}\right\}$ is the matrix of the STAR-RIS's coefficients with $\kappa \in \{{\rm r}, {\rm t}\}$ representing the reflected or transmitted coefficients of the STAR-RIS, where the amplitudes $\beta_\kappa^m[n]$ and phases $\phi_\kappa^m[n]$ of the STAR-RIS should satisfy $(\beta_{\rm r}^m[n])^2 + (\beta_{\rm t}^m[n])^2 = 1$, $\phi_{\rm r}^m[n], \phi_{\rm t}^m[n] \in [0, 2\pi),\ \forall m \in \mathcal{M}$.

It is assumed that the ground users are with limited resources for local computing and thus users' tasks have to be offloaded to the BS and UAV for computing. Let $l_k^{\rm BS}[n]$ and $l_k^{\rm ua}[n]$ denote the number of the offloaded bits that need to be computed at the BS and UAV for user $k$ in the $n$-th time slot, respectively. Due to the fact that the BS and UAV can only deal with the tasks they have received, the following constraints should be satisfied

$$\delta_{\rm t} \zeta_k[n] R_k^{\rm ua}[n] \geq l_k^{\rm ua}[n],\ \forall k \in \mathcal{K}, n \in \mathcal{N}, \tag{5}$$

$$\delta_{\rm t} \zeta_k[n] R_k^{\rm BS}[n] \geq l_k^{\rm BS}[n],\ \forall k \in \mathcal{K}, n \in \mathcal{N}. \tag{6}$$

In order to ensure that every user's minimum computational need is met, we implement the following QoS constraints

$$\sum_{n=1}^{N} \left(l_k^{\rm BS}[n] + l_k^{\rm ua}[n]\right) \geq L_{\rm tol}, \sum_{n=1}^{N} l_k^{\rm BS}[n] \geq L_{\rm b},\ \forall k \in \mathcal{K}, \tag{7}$$

where $L_{\rm tol}$ is the minimum total computing task requirement for each user, $L_{\rm b}$ denotes the the minimum computing task need for each user at BS.

Let $f_k^{\rm BS}[n]$ and $f_k^{\rm ua}[n]$ respectively denote the computation frequency allocated by the BS and UAV for the $k$-th user at the $n$-th time slot, we have

$$\sum_{k=1}^{K} f_k^{\rm BS}[n] \leq F_{\rm BS},\ \sum_{k=1}^{K} f_k^{\rm ua}[n] \leq F_{\rm ua},\ \forall n \in \mathcal{N}, \tag{8}$$

where $F_{\rm BS}$ and $F_{\rm ua}$ are the maximum CPU frequency at the BS and the UAV, respectively. In addition, considering the processing causality, it is assumed that the BS and the UAV solely receive the offloaded tasks without carrying out any computations within the first time slot, and the users cease the act of offloading their tasks at the last time slot, i.e., $f_k^{\rm BS}[1] = f_k^{\rm ua}[1] = l_k^{\rm BS}[N] = l_k^{\rm ua}[N] = 0$ for $k \in \mathcal{K}$. In order to guarantee that all users' offloaded task-input data can be completely computed within the give mission time $T$, we have the following information-causality constraints:

$$\sum_{i=2}^{n} \frac{f_k^{\rm ua}[i]\delta_{\rm t}}{\varrho_{\rm ua}} \geq \sum_{i=1}^{n-1} l_k^{\rm ua}[i],\ \forall k \in \mathcal{K}, n \in \mathcal{N}_1, \tag{9}$$

$$\sum_{i=2}^{n} \frac{f_k^{\rm BS}[i]\delta_{\rm t}}{\varrho_{\rm BS}} \geq \sum_{i=1}^{n-1} l_k^{\rm BS}[i],\ k \in \mathcal{K}, \forall n \in \mathcal{N}_1, \tag{10}$$

where $\varrho_{\rm BS}$ and $\varrho_{\rm ua}$ represent the number of CPU cycles needed for processing 1-bit of task-input data at the BS and UAV, respectively, and $\mathcal{N}_1 \triangleq \{2, 3, \cdots, N\}$ is a subset of $\mathcal{N}$. Therefore, the total amount of the completed task-input data for all users within the whole period can be calculated as $L_{\rm tol} = \sum_{n=1}^{N} \sum_{k=1}^{K} l_k^{\rm BS}[n] + l_k^{\rm ua}[n]$, which is an important indicator to measure the computing capability of the MEC system.

## III. Problem Formulation and Algorithm Design

### A. Optimization Problem Formulation

In this section, the optimization problem will be formulated based on the analysis in Section II. In particular, we try to maximize the sum offloading tasks for all users, while ensuring the QoS constraints of users with minimum computational requirements, by jointly optimizing the resource allocation variable $\mathbf{L} \triangleq \{l_k^{\rm ua}[n], l_k^{\rm BS}[n], f_k^{\rm ua}[n], f_k^{\rm BS}[n], k \in \mathcal{K}, n \in \mathcal{N}\}$, user scheduling variable $\boldsymbol{\zeta} \triangleq \{\zeta_k[n], k \in \mathcal{K}, n \in \mathcal{N}\}$, passive beamforming variable $\mathbf{\Upsilon} \triangleq \{\mathbf{\Theta}_{\rm r}[n], \mathbf{\Theta}_{\rm t}[n], n \in \mathcal{N}\}$ and UAV trajectory variable $\mathbf{Q} \triangleq \{\mathbf{q}_{\rm ua}[n], n \in \mathcal{N}\}$. Hence, the corresponding optimization problem can be formulated as

$$\max_{\mathbf{L}, \boldsymbol{\zeta}, \mathbf{\Upsilon}, \mathbf{Q}} \quad L_{\rm tol},$$

$$\text{s.t.} \quad (5) - (10), \tag{11a}$$

$$\|\mathbf{v}_{\rm ua}[n]\| \leq v_{\max}, \|\mathbf{a}_{\rm ua}[n]\| \leq a_{\max},\ \forall n \in \mathcal{N}, \tag{11b}$$

$$\mathbf{q}_{\rm ua}[0] = \mathbf{q}_{\rm I}, \mathbf{q}_{\rm ua}[N] = \mathbf{q}_{\rm F}, \tag{11c}$$

$$\sum_{k=1}^{K} \zeta_k[n] = 1, \zeta_k[n] \in \{0, 1\},\ n \in \mathcal{N}, k \in \mathcal{K}, \tag{11d}$$

$$f_k^{\rm ua}[1] = f_k^{\rm BS}[1] = l_k^{\rm ua}[N] = l_k^{\rm BS}[N] = 0,\ \forall k \in \mathcal{K}, \tag{11e}$$

$$(\beta_{\rm r}^m[n])^2 + (\beta_{\rm t}^m[n])^2 = 1,\ \forall m \in \mathcal{M}, n \in \mathcal{N}, \tag{11f}$$

$$\beta_{\rm r}^m[n], \beta_{\rm t}^m[n] \in (0, 1),\ \forall m \in \mathcal{M}, n \in \mathcal{N}, \tag{11g}$$

$$\phi_{\rm r}^m[n], \phi_{\rm t}^m[n] \in [0, 2\pi),\ \forall m \in \mathcal{M}, n \in \mathcal{N}. \tag{11h}$$

Actually, the problem (11) is a non-convex problem due to the non-convexity of constraints (5), (6) and (11f), which is difficult to solve directly. To overcome this challenge, the alternative strategy is employed to divide the optimization problem (11) into three subproblems. The algorithm is designed by alternatively optimizing three variable subsets, which are respectively denoted as $\mathbf{\Xi}_1 = \{\mathbf{L}, \boldsymbol{\zeta}\}$, $\mathbf{\Xi}_2 = \{\mathbf{L}, \mathbf{\Upsilon}\}$ and $\mathbf{\Xi}_3 = \{\mathbf{L}, \mathbf{Q}\}$. More details of the algorithm design is given in the next subsection.

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### B. Algorithm Design

*1) Designing $\boldsymbol{\Xi}_1$ with the given $\mathbf{Q}$ and $\boldsymbol{\Upsilon}$:* First, we jointly optimize the resource allocation variable $\mathbf{L}$ and user scheduling variable $\boldsymbol{\zeta}$ with the given passive beamforming and UAV trajectory. In this case, the original problem (11) can be simplified as

$$\max_{\boldsymbol{\Xi}_1} \ L_{\mathrm{tol}}(\boldsymbol{\Xi}_1), \quad \text{s.t. (11a), (11d), (11e).} \tag{12a}$$

It is important to highlight that the optimization problem (12) is a non-convex problem because of the binary variable $\boldsymbol{\zeta}$. To address this optimization problem, the non-convex binary constraint (11d) is equivalently transformed as the following constraints:

$$\sum_{k=1}^{K} \zeta_k[n] = 1, 0 \le \zeta_k[n] \le 1, \ \forall n \in \mathcal{N}, k \in \mathcal{K}, \tag{13}$$

$$\eta_k[n] = \zeta_k[n] - \zeta_k^2[n] = 0, \ \forall n \in \mathcal{N}, k \in \mathcal{K}. \tag{14}$$

Note that for any $\zeta_k[n] \in [0,1]$, $\eta_k[n] \ge 0$ always holds and the equality in (13) is satisfied if and only if $\zeta_k[n] = 0$ or $\zeta_k[n] = 1$. Considering the non-negative characteristic of $\{\eta_k[n]\}_{k\in\mathcal{K},n\in\mathcal{N}}$, we try to add the sum of them into the objective function as a penalty term that is subtracted from the objective function. To guarantee the fulfilment of the binary constraint (11d), an additional inner loop iteration is incorporated into the $(l+1)$-th iteration of the outer loop to iteratively enforce the penalty term approaching to 0. Note that the inclusion of the penalty term in the objective function results in a non-concave objective function due to the convex nature of $-\eta_k[n]$ with respect to (w.r.t.) $\zeta_k[n]$. To handle this problem, we utilize the liner upper bound, i.e., the first-order Taylor expansion of $\eta_k[n]$, to replace itself. The liner upper bound of $\eta_k[n]$ in the $(t+1)$-th inner loop iteration can be expressed as

- $\eta_k[n] \le \widehat{\eta}_k[n]\big(\zeta_k[n], \zeta_k^{(t)}[n]\big) = \zeta_k[n] - \big((\zeta_k^{(t)}[n])^2 + 2\zeta_k^{(t)}[n](\zeta_k[n] - \zeta_k^{(t)}[n])\big)$.

Thus, in the $(t+1)$-th inner loop iteration of the $(l+1)$-th iteration of the outer loop, the optimization problem (12) can be re-expressed

$$\max_{\boldsymbol{\Xi}_1} \ L_{\mathrm{tol}}(\boldsymbol{\Xi}_1) - \hat{\rho}\sum_{n=1}^{N}\sum_{k=1}^{K}\widehat{\eta}_k[n]\left(\zeta_k[n], \zeta_k^{(t)}[n]\right), \quad \text{s.t.} \ \ (11a), (11e), (13), \tag{15a}$$

where $\hat{\rho} > 0$ denotes the introduced penalty coefficient. The problem (15) is a convex optimization problem and can be directly solved by the existing tool such as CVX.

*2) Designing $\boldsymbol{\Xi}_2$ with given $\boldsymbol{\zeta}$ and $\mathbf{Q}$:* After achieving the user scheduling $\boldsymbol{\zeta}$, we focus on designing the passive beamforming variables with the given UAV's trajectory. In particular, the corresponding optimization problem for passive beamforming can be expressed as

$$\max_{\boldsymbol{\Xi}_2} \ \ L_{\mathrm{tol}}(\boldsymbol{\Xi}_2), \quad \text{s.t. } (11a), (11e) - (11h). \tag{16a}$$

Note that, problem (16) is a non-convex optimization problem because of the non-convexity of constraints (5), (6) and (11f). However, we can derive the close-form expression of the optimal reflected phases based on the maximum-ratio transmission (MRT), the obtained optimal reflection phases at the $n$-th time slot can be given by

$$\boldsymbol{\phi}_{\mathrm{r}}^{\mathrm{opt}}[n] = \arg(\mathrm{norm}(\mathbf{h}_{\mathrm{rb}}^{*}[n] \circ \mathbf{h}_{\mathrm{r}k}[n])), \tag{17}$$

where the selection of the index $k$ is determined by the condition of $\zeta_k[n] = 1$.

Next, we will focus on designing the reflection amplitudes and transmitted coefficient of STAR-RIS's passive beamforming, i.e., $\{\beta_{\mathrm{r}}^m[n]\}_{m\in\mathcal{M},n\in\mathcal{N}}$ and $\{\boldsymbol{\Theta}_{\mathrm{t}}[n]\}_{n\in\mathcal{N}}$. For the non-convex constraints (5) and (6), we first rewrite $\boldsymbol{\vartheta}_{\mathrm{r}}[n] = \boldsymbol{\Phi}_{\mathrm{r}}[n]\boldsymbol{\beta}_{\mathrm{r}}[n]$, where $\boldsymbol{\vartheta}_{\mathrm{r}}[n] = \mathrm{diag}(\boldsymbol{\Theta}_{\mathrm{r}}[n])$, $\boldsymbol{\beta}_{\mathrm{r}}[n] = \{\beta_{\mathrm{r}}^1[n], \cdots, \beta_{\mathrm{r}}^m[n], \cdots, \beta_{\mathrm{r}}^M[n]\}^T$, and $\boldsymbol{\Phi}_{\mathrm{r}}[n] = \mathrm{Diag}(e^{j\boldsymbol{\phi}_{\mathrm{r}}^{\mathrm{opt}}[n]})$. Therefore, $R_k^{\mathrm{BS}}[n]$ and $R_k^{\mathrm{ua}}[n]$ can be further re-expressed as

- $R_k^{\mathrm{BS}}[n] = \zeta_k[n]B\log_2\left(1 + \boldsymbol{\beta}_{\mathrm{r}}^T[n]\mathbf{F}_k[n]\boldsymbol{\beta}_{\mathrm{r}}[n]\right)$,
- $R_k^{\mathrm{ua}}[n] = \zeta_k[n]B\log_2\left(1 + \boldsymbol{\vartheta}_{\mathrm{t}}^T[n]\mathbf{E}_k[n]\boldsymbol{\vartheta}_{\mathrm{t}}[n]\right)$,

where

- $\boldsymbol{\vartheta}_{\mathrm{t}}[n] = \mathrm{diag}(\boldsymbol{\Theta}_{\mathrm{t}}[n]) = \{\beta_{\mathrm{t}}^1[n]e^{j\phi_{\mathrm{t}}^1[n]}, \cdots, \beta_{\mathrm{t}}^m[n]e^{j\phi_{\mathrm{t}}^m[n]}, \cdots, \beta_{\mathrm{t}}^M[n]e^{j\phi_{\mathrm{t}}^M[n]}\}$,
- $\mathbf{F}_k[n] = \frac{\boldsymbol{\Phi}_{\mathrm{r}}^H[n]}{\sigma_{\mathrm{BS}}^2}(\mathbf{h}_{\mathrm{BR}}^{*}[n] \circ \mathbf{h}_{\mathrm{r}k}[n])(\mathbf{h}_{\mathrm{BR}}^{*}[n] \circ \mathbf{h}_{\mathrm{r}k}[n])^H\boldsymbol{\Phi}_{\mathrm{r}}$,
- $\mathbf{E}_k[n] = (\mathbf{h}_{\mathrm{ru}}^{*} \circ \mathbf{h}_{\mathrm{r}k}[n])(\mathbf{h}_{\mathrm{ru}}^{*} \circ \mathbf{h}_{\mathrm{r}_k}[n])^H$.

Then, we introduce auxiliary variables $\gamma_k^{\mathrm{BS}}[n]$ and $\gamma_k^{\mathrm{ua}}[n]$ which satisfy $\gamma_k^{\mathrm{BS}}[n] \le \boldsymbol{\beta}_{\mathrm{r}}^T[n]\mathbf{F}_k[n]\boldsymbol{\beta}_{\mathrm{r}}[n]$ and $\gamma_k^{\mathrm{ua}}[n] \le \boldsymbol{\vartheta}_{\mathrm{t}}^H[n]\mathbf{E}_k[n]\boldsymbol{\vartheta}_{\mathrm{t}}[n]$. Thus, the problem designing reflection amplitudes and transmission coefficients in the $(l+1)$-th iteration of the proposed algorithm can be equivalently transformed as

$$\max_{\boldsymbol{\beta}_{\mathrm{r}},\boldsymbol{\vartheta}_{\mathrm{t}},\boldsymbol{\gamma}} \ L_{\mathrm{tol}}(\boldsymbol{\beta}_{\mathrm{r}}, \boldsymbol{\vartheta}_{\mathrm{t}}, \boldsymbol{\gamma})$$

$$\text{s.t. } (7) - (10), (11e), \tag{18a}$$

$$\delta_{\mathrm{t}}[n]\zeta_k[n]\log_2(1 + \gamma_k^{\mathrm{ua}}[n]) \ge l_k^{\mathrm{ua}}[n], \ \forall k \in \mathcal{K}, n \in \mathcal{N}, \tag{18b}$$

$$\delta_{\mathrm{t}}[n]\zeta_k[n]\log_2(1 + \gamma_k^{\mathrm{BS}}[n]) \ge l_k^{\mathrm{BS}}[n], \forall k \in \mathcal{K}, n \in \mathcal{N}, \tag{18c}$$

$$\gamma_k^{\mathrm{ua}}[n] \le \boldsymbol{\vartheta}_{\mathrm{t}}^H[n]\mathbf{E}_k[n]\boldsymbol{\vartheta}_{\mathrm{t}}, \ \forall k \in \mathcal{K}, n \in \mathcal{N}, \tag{18d}$$

$$\gamma_k^{\mathrm{BS}}[n] \le \boldsymbol{\beta}_{\mathrm{r}}^T[n]\mathbf{F}_k[n]\boldsymbol{\beta}_{\mathrm{r}}[n], \ \forall k \in \mathcal{K}, n \in \mathcal{N}, \tag{18e}$$

$$(\beta_{\mathrm{r}}^m[n])^2 + (\beta_{\mathrm{t}}^m[n])^2 \le 1, \ \forall m \in \mathcal{M}, n \in \mathcal{N}. \tag{18f}$$

where $\boldsymbol{\gamma} \triangleq \{\gamma_k^{\mathrm{ua}}[n], \gamma_k^{\mathrm{BS}}[n], k \in \mathcal{K}, n \in \mathcal{N}\}$. Actually, problem (18) is still a non-convex optimization problem due to the non-convex constraints (18d) and (18e). In order to address this issue, we employ a linear lower bound, specifically the first-order Taylor expansion, to approximate the right-hand side of constraints (18d) and (18e) and replace them accordingly. It is worth noting that the equality sign in constraint (11f) has been substituted with an inequality sign to establish the convex constraint (18f), transforming (18) into a convex optimization problem. In fact, this replacement of the equal sign does not impact the fulfilment of the equality constraint, because the

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constraint (18f) is satisfied with strict equality in the optimal solution of problem (18).

*3) Designing $\mathbf{\Xi}_3$ with given $\boldsymbol{\zeta}$ and $\boldsymbol{\Upsilon}$:* Next, we will design the trajectory of the UAV with the obtained $\boldsymbol{\zeta}$ and $\boldsymbol{\Upsilon}$. Specifically, the corresponding optimization problem for UAV trajectory can be expressed as

$$\max_{\mathbf{\Xi}_3} \; L_{\mathrm{tol}}(\mathbf{\Xi}_3), \quad \text{s.t. (11a)} - \text{(11c)}, \text{(11e)}. \tag{19a}$$

In fact, problem (19) is non-convex because of constraints (5) and (6). In order to handle this non-convex constraints, we first introduce the auxiliary variables $\lambda_k[n]$ and $\tilde{\lambda}[n]$ with $\lambda_k[n] \geq d_{\mathrm{r}k}^{\alpha_{\mathrm{r}k}}[n] = \|\mathbf{q}_{\mathrm{ua}}[n] - \mathbf{q}_k\|^{\alpha_{\mathrm{r}k}}$, $\tilde{\lambda}[n] \geq d_{\mathrm{rb}}^{\alpha_{\mathrm{rb}}}[n] = \|\mathbf{q}_{\mathrm{BS}} - \mathbf{q}_{\mathrm{ua}}[n]\|^{\alpha_{\mathrm{rb}}}$. Hence, we have

- $R_k^{\mathrm{ua}}[n] \geq \tilde{R}_k^{\mathrm{ua}}[n] = \log_2\left(1 + \frac{\rho p \left|\mathbf{h}_{\mathrm{ru}}^H \mathbf{\Theta}_{\mathrm{t}}^H[n] \widehat{\mathbf{h}}_{\mathrm{u}k}[n]\right|^2}{\lambda_k[n]\sigma_{\mathrm{ua}}^2}\right)$,
- $R_k^{\mathrm{BS}}[n] \geq \tilde{R}_k^{\mathrm{BS}}[n] = \log_2\left(\frac{\rho^2 p_k[n] \left|\widehat{\mathbf{h}}_{\mathrm{BR}}^H[n] \mathbf{\Theta}_{\mathrm{r}}^H[n] \widehat{\mathbf{h}}_{\mathrm{u}k}[n]\right|^2}{\lambda_k[n]\tilde{\lambda}[n]\sigma_{\mathrm{ua}}^2} + 1\right)$.

Actually, $\tilde{R}_k^{\mathrm{ua}}[n]$ is a convex function w.r.t. $\lambda_k[n]$ and $\tilde{R}_k^{\mathrm{BS}}[n]$ is the jointly convex function w.r.t. $\lambda_k[n]$ and $\tilde{\lambda}[n]$, and thus we can apply the first-order Taylor expansion of $\tilde{R}_k^{\mathrm{ua}}[n]$ and $\tilde{R}_k^{\mathrm{BS}}[n]$ at the given point $(\lambda_k^{(l)}[n], \tilde{\lambda}^{(l)}[n])$ in $(l+1)$-th iteration of the proposed algorithm to convert constraints (5) and (6) as the following convex constraints

$$\zeta_k[n]\delta_{\mathrm{t}}\widehat{R}_k^{\mathrm{ua}}[n] \geq l_k^{\mathrm{ua}}[n], \; \forall k \in \mathcal{K}, n \in \mathcal{N}, \tag{20}$$

$$\zeta_k[n]\delta_{\mathrm{t}}\widehat{R}_k^{\mathrm{BS}}[n] \geq l_k^{\mathrm{BS}}[n], \; \forall k \in \mathcal{K}, n \in \mathcal{N}, \tag{21}$$

where $\widehat{R}_k^{\mathrm{ua}}[n]$ and $\widehat{R}_k^{\mathrm{BS}}[n]$ denotes the first-order Taylor expansion of $\tilde{R}_k^{\mathrm{ua}}[n]$ and $\tilde{R}_k^{\mathrm{BS}}[n]$ at the given point $(\lambda_k^{(l)}[n], \tilde{\lambda}^{(l)}[n])$, respectively.

As a result, the optimization problem (19) can be transformed as

$$\max_{\mathbf{\Xi}} \quad \mathbf{L}(\mathbf{\Xi}),$$

$$\text{s.t. (7)} - \text{(10)}, \text{(11b)}, \text{(11c)}, \text{(11e)}, (20), (21), \tag{22a}$$

$$\lambda_k[n] \geq \|\mathbf{q}_{\mathrm{ua}}[n] - \mathbf{q}_k\|^{\alpha_{\mathrm{r}k}}, \tag{22b}$$

$$\tilde{\lambda}[n] \geq \|\mathbf{q}_{\mathrm{BS}} - \mathbf{q}_{\mathrm{ua}}[n]\|^{\alpha_{\mathrm{rb}}}, \tag{22c}$$

where $\mathbf{\Xi} = \{\mathbf{\Xi}_3, \lambda_k[n], \tilde{\lambda}[n], \mu[n]\}$. Consequently, we can leverage the convex optimization solver, e.g., CVX, to address this problem.

## IV. Simulation Results

To highlight the effectiveness of the proposed STAR-RIS-aided UAV-enabled MEC scheme, we present the numerical simulation results in this section and compare the their with three benchmark schemes, including: **1) RIS-aided scheme [15]:** In this baseline scheme, two adjacent conventional RISs with $\frac{M}{2}$ elements are adopted to replace the STAR-RIS, where one is the reflecting-only RIS and the other one is the transmission-only RIS; **2) Without trajectory optimization scheme:** This scheme focuses on optimizing $\mathbf{L}$, $\boldsymbol{\zeta}$ and $\boldsymbol{\Upsilon}$ with direct UAV trajectory flying from the initial point to the final point at a consistent speed. **3) Heuristic scheme:** UAV will traverse each user node based on the pre-defined trajectory at a consistent speed. Similarly, variables $\mathbf{L}$, $\boldsymbol{\zeta}$ and $\boldsymbol{\Upsilon}$ will be optimized in this scheme. In addition, the simulation parameters are listed as $H = 30$ m, $B = 1$ MHz, $\iota_{\mathrm{ua}} = \iota_{\mathrm{BS}} = 10^{-7}$, $\lambda_{\mathrm{c}} = 2.4$ GHz, $\mathbf{q}_{\mathrm{I}} = [-40, 0, 30]^T$m, $\mathbf{q}_{\mathrm{F}} = [40, 0, 30]^T$m, $v_{\max} = 30$ m/s, $a_{\max} = 50$ m/$s^2$, $F_{\mathrm{BS}} = 20$ GHz, $\sigma_{\mathrm{ua}}^2 = \sigma_{\mathrm{BS}}^2 = -100$ dbm, $\alpha_{\mathrm{r}k} = \alpha_{\mathrm{rb}} = 2.3$, $\varrho_{\mathrm{ua}} = \varrho_{\mathrm{BS}} = 1000$ cycles/bit, $\delta_{\mathrm{t}} = 0.3$ s.

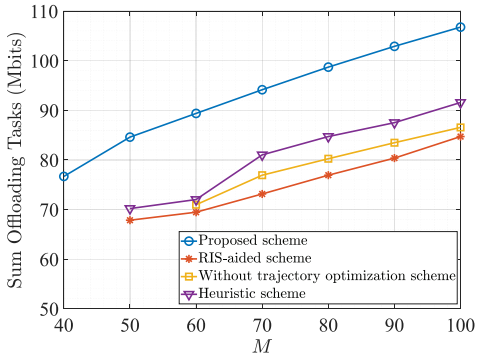



Fig. 2. Sum offloading tasks versus the number of elements at STAR-RIS with $L_{\mathrm{tol}} = 10$ Mbits, $L_{\mathrm{b}} = 2$ Mbits, $N = 50$ and $F_{\mathrm{ua}} = 4$ GHz.

In Fig. 2, we investigate how the quantity of elements at STAR-RIS $M$ impacts the sum offloading data for individual users, taking into account $L_{\mathrm{tol}} = 10$ Mbits, $L_{\mathrm{b}} = 2$ Mbits, $N = 50$, and $F_{\mathrm{ua}} = 4$ GHz. Specifically, it is observed that the total offloading data shows a positive correlation with the increase of $M$ for all cases, as the more elements can offer more spatial degrees of freedom (DoFs) to modify the wireless propagation environment. Additionally, it is noted that the baseline schemes face challenges with incomplete data when the value of $M$ is lower. This is because a smaller $M$ makes it difficult for these schemes to fulfil the minimum computing task requirement for each user at the BS ($L_{\mathrm{b}}$), especially when accounting for the two-path signal loss from users to the BS. This shows the application potential of the proposed MEC scheme in balancing resource allocation. Note that the traditional RIS-assisted system shows the smallest performance improvement, whereas the proposed scheme aided by the STAR-RIS achieves the most substantial performance enhancement. This indicates that STAR-RIS demonstrates a significant potential in enhancing the computing capability of the MEC networks.

The influence of the number of time slots $N$ on the sum offloading data for all users is explored by considering $L_{\mathrm{tol}} = 10$ Mbits, $L_{\mathrm{b}} = 2$ Mbits, $M = 50$ and $F_{\mathrm{ua}} = 4$ GHz, as shown in Fig. 3. In particular, the computational capacity of the MEC system exhibits a consistent upward trend across all scenarios as the variable $N$ increases. Similarly, the performance data for these baseline schemes is incomplete as a result of the challenges in meeting the QoS requirements stipulated in equation (7) caused by the lower values of $N$ and $M$. Furthermore, the suggested MEC scheme continues to demonstrate superior performance improvement compared to the traditional RIS-assisted MEC scheme, which exhibits the least enhancement in performance.

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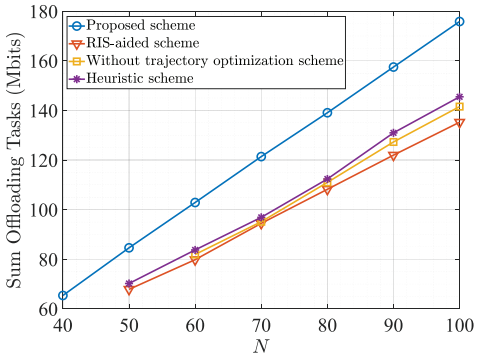



Fig. 3. Sum offloading tasks versus the number of time slots with $L_{\rm tol} = 10$ Mbits, $L_{\rm b} = 2$ Mbits, $M = 50$ and $F_{\rm ua} = 4$ GHz.

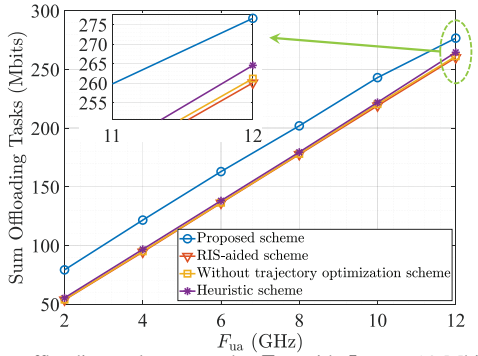



Fig. 4. Sum offloading tasks versus the $F_{\rm ua}$ with $L_{\rm tol} = 10$ Mbits, $L_{\rm b} = 2$ Mbits, $N = 70$ and $M = 50$.

Fig.4 presents the varied correlation between the maximum CPU frequency at the UAV $F_{\rm ua}$ and the total amount of offloading tasks when $L_{\rm tol}$, $L_{\rm b}$, $N$ and $M$ are set to 10 Mbits, 2 Mbits, 70 and 50, respectively. From this picture, we can find that the sum of users' offloading tasks increases linearly as $F_{\rm ua}$, which is because the UAV can compute the more computational tasks from users with the growth of $F_{\rm ua}$. In addition, it is worth noting that STAR-RIS supported schemes consistently achieve superior performance gains compared to conventional RIS-assisted schemes, even in cases where optimized DoFs are not utilized, such as the scheme without trajectory optimization or Heuristic scheme, which further demonstrates the superior of STAR-RIS in enhancing the computational capability of the MEC networks.

## V. Conclusion

In this paper, we propose a MEC scheme assisted by STAR-RIS and UAV, which allows the bi-directional task offloading so that users can offload their tasks to the MEC servers situated at the BS and UAV simultaneously. Then, a non-convex optimization problem is established which seeks to maximize the sum of users' offloading tasks while guaranteeing the QoS constraints. To effectively address this non-convex optimization problem, we propose an iterative algorithm that draws inspiration from the SCA technique. The efficacy of the proposed MEC scheme is substantiated through the simulation outcomes in comparison with the several baseline schemes, encompassing the traditional RIS-assisted scheme.

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